 

Economic & Policy Priorities for Carbon Management Workshop Summary

LBJ Center, Washington, DC | April 29, 2025

Workshop Overview

The workshop aimed to establish priorities for the deployment of large-scale carbon management in the U.S., with a focus on CCUS, DAC, carbon hubs, and related policy issues. Discussions addressed federal incentives, legal and regulatory barriers, and environmental justice within the context of the current administration.

Session 1: §45Q and the Design & Implementation of Federal Incentives for Carbon Management in the New Administration

This session examined the 45Q tax credit and the design of federal incentives. Key themes included:

- The economic realities of carbon capture, noting the limited private market interest outside of enhanced oil recovery (EOR), and the necessity of public incentives to drive deployment. The continuation of capture is dependent on the continuation of incentives, and infrastructure development is tied to anticipated future revenues. A sustained, increasing subsidy may have differential impacts over the short- versus long-run, namely shielding the industry from market competition.
- Analysis of the costs and benefits of retrofitting fossil fuel power plants with carbon capture, utilizing an integrated assessment model to estimate capital and operating costs of capture, transport and storage. The social benefits of carbon capture and storage are evaluated by considering the reduction in damages from both carbon dioxide emissions and local air pollution. The modeling suggests that significant investments are incentivized by 45Q and air quality benefits from CCUS are more significant in populated areas.
- Policy proposals to enhance 45Q's effectiveness, including adjusting the credit for inflation to maintain its value, increasing credit values to better align with capture

costs, and providing equivalent support for carbon capture and utilization (CCU) projects to ensure fair competition. The impact of inflation and rising borrowing costs on diminishing the value of the 45Q tax credit was discussed.

Session 2: Legal and Regulatory Barriers to CCUS Adoption

This session addressed the legal and regulatory obstacles to CCUS deployment. Key themes included:

- Identification of barriers such as incomplete regulatory regimes, inadequate monitoring, reporting, and verification (MRV) frameworks, and challenges in permitting, particularly for Class VI wells. The slow development of pipeline infrastructure was highlighted as a bottleneck.
- Policy recommendations to overcome these barriers, such as expanding support for Class VI well permitting and state primacy, reforming federal permitting processes to reduce timelines, and establishing clear guidelines for pipeline development, including open-access service and rate regulation requirements.
- The scaling of CCUS projects in the US, noting the growth in operational and announced projects, but also highlighting warning signs related to the US share of global projects.

Session 3: Community Impacts of Carbon Management

This session focused on the community impacts of carbon management, emphasizing environmental justice. Key themes included:

- The importance of considering equity and distributional impacts of CCUS projects, acknowledging both positive externalities like GHG reduction and job creation, and negative externalities like potential seismic activity and groundwater contamination. Public perception and skepticism towards CCUS were also acknowledged. The necessity of prioritizing community engagement and co-design in DAC project deployment to ensure regionally appropriate co-benefits and address potential injustices. This includes understanding each community's vision for procedural, distributive, reparative, and transformative justice. The challenges in cultivating support from environmental justice groups and frontline communities, and the importance of supporting communities that are explicitly requesting and planning for DAC projects.
- Research leveraging spatial and temporal variations of CCUS projects and individual housing transaction data to estimate how potential economic, environmental and geological impacts are reflected in the housing market using hedonic regression

estimates. This research aims to reflect the net impacts of the benefits and costs that CCUS projects can bring to local communities.

- Analysis using a power systems model of the U.S. compared existing incentives to a cap and trade policy targeting 72% CO_2 emission reduction. Current policies (i.e. 45Q) incentivize substantial coal CCUS adoption and a small amount of new CCUS natural gas plants. Cap and trade would result in substantially more conversion. Analysis of air quality impacts indicates that retrofitting power plants with carbon capture technology would change the generation reduce ambient air pollution.

Resources:

UT-Austin/UWyo Economics of CCUS Project

Slides, Chuck Mason, University of Wyoming

Slides, Emily Pope, C2ES

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